Pressure Measurement

For most applications, we usually do not want to measure to actual/absolute pressure of a fluid, as it includes atmospheric pressure.

We usually care more about how the pressure differs from atmospheric, for example, a tire having atmospheric pressure inside is not that useful to know, but knowing that the difference between the two is 0, it gives us more useful information; i.e. the tire is deflated.

Thus, we use gauge and vacuum pressure instead, which measures relative to atmospheric.

Gauge & Vacuum Pressure

Gauge Pressure is measured as

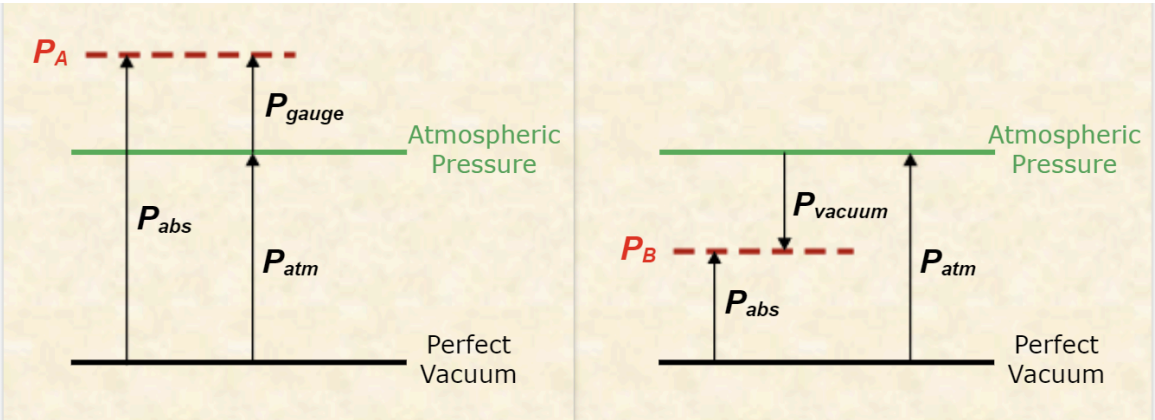
$$P_{gauge} + P_{atm} = P_{abs}$$

In other words, gauge pressure is the extra pressure compared to atmospheric.

Vacuum Pressure is measured as

$$P_{atm} - P_{abs} = P_{vac}$$

which is just how below the pressure is from atmospheric.



For example, if we had a pressure meter (like a bourdon gauge) reading 0 gauge pressure has an absolute pressure of 1 atm, or about 1.013 bar

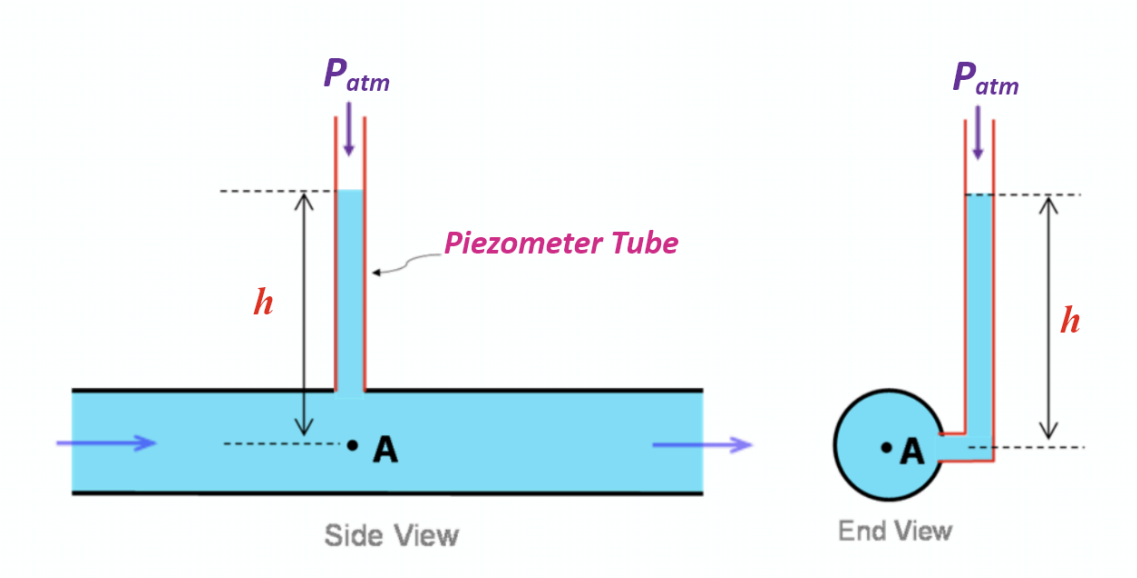
A meter reading 5 bars gauge pressure has an absolute pressure of 5+1.013=6.013 bars, since we have to account for atmospheric.

When a meter reads -1 bar gauge pressure, it has 1 vacuum pressure, or 1.013 - 1 =0.013 bar absolute pressure.

Generally, most pressure measurement systems output gauge or vacuum pressure, and most questions that do not state what atmospheric pressure is want pressure as gauge or vacuum pressure.

Piezometer

A piezometer is a open vertical tube usually connected to a pipe with flowing fluid, which allows fluid to flow into the tube.



The piezometer helps to measure the (static pressure? i think?) at point A.

The liquid flows into the tube, and will equalise at a certain point, when the atmospheric pressure downwards is equal to the fluid's pressure pushing upwards.

Thus, at point A, the gauge pressure is just the pressure due to the liquid column:

$$P_{\text{gauge}} = \rho g h$$

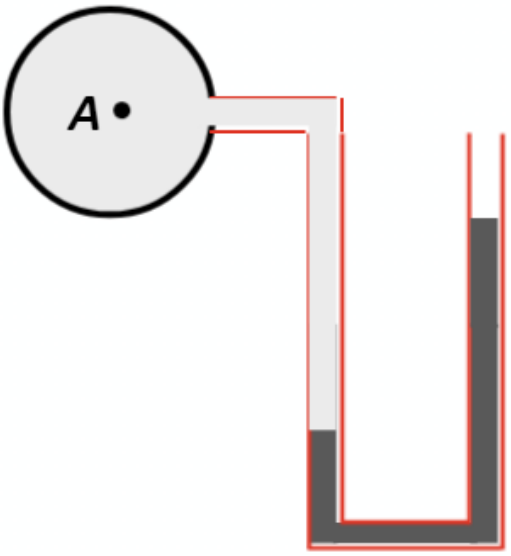
and the absolute pressure will include the atmospheric pressure

$$P_{\text{abs}} = P_{atm} + \rho g h$$

Piezometers are quite useful as they are simple to operate and accurate, however only work when the pressure is greater than atmospheric, otherwise gas will be sucked in, and is only practical for small gauge pressure, to prevent a very long piezometer tube. Piezometers also only work with liquids, gas pressure cannot be measured like this.

U-Tube Manometer

A U tube Manometer is a U-shaped glass tube, with a gauge fluid inside that is heavier than the fluid



Usually, one end is exposed to atmospheric while the other is connected to a gas/liquid pipe.

Generally, the (absolute) pressure of A can be found by forming an equation with the pressure at two points at equal height at each arm of the manometer.

There are 2 factors to consider, whether the gauge fluid is more on the left or right side, and if the fluid is a gas or liquid.

In this case, the fluid in the pipe is a gas, thus the hydrostatic pressure of the fluid column on the left arm is negligable, and we can say that Pa = Pb on the left case.

When the fluid pushes more on the gauge fluid that atmospheric, its absolute pressure is

$$P_{abs} = P_{atm} + \rho gh$$

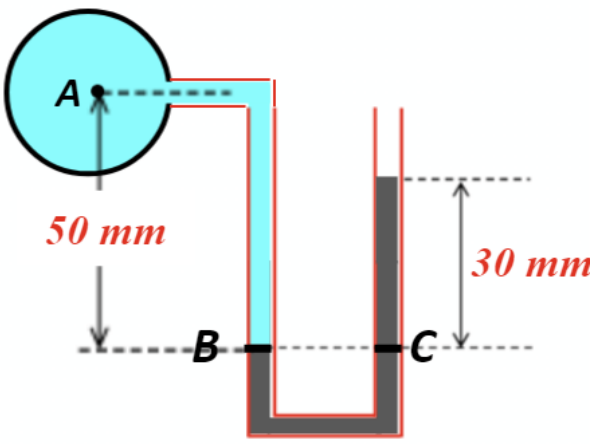
where rho g h represents the hydrostatic pressure of the gauge fluid's liquid column.

Thus, for its gauge pressure (as it pushes more than atmospheric). its the same as the hydrostatic pressure of the gauge fluid's liquid column.

When the atmospheric pressure pushes more than the fluid, its absolute pressure is

$$P_{abs} + \rho gh = P_{atm}$$
$$P_{abs} = P_{atm} - \rho gh$$

Since atmospheric pressure is higher than the fluid pressure, we use vacuum pressure instead, which is the same as the gauge fluid's liquid column.



For liquids, the hydrostatic pressure due the liquid on the left arm is no longer negligible, thus the absolute pressure at point B is

$$P_A + \rho g h$$

Pa is essentially the static pressure of the liquid in the pipe (i think)

So for the above case, absolute pressure of A is

$$P_{abs} + \rho_1 g h_1 = P_{atm} + \rho_2 g h_2$$

and to find gauge pressure, (sub in gauge and absolute pressure relationship)

$$(P_{gauge} + \cancel{P_{atm}}) + \rho_1 g h_1 = \cancel{P_{atm}} + \rho_2 g h_2$$

$$P_{gauge} + \rho_1 g h_1 = \rho_2 g h_2$$

The same concept can be applied for when the fluid has a lower pressure than atmospheric.

However, usually you can assume Pa to have gauge pressure, then when the value is negative, swap to vacuum pressure at the end.

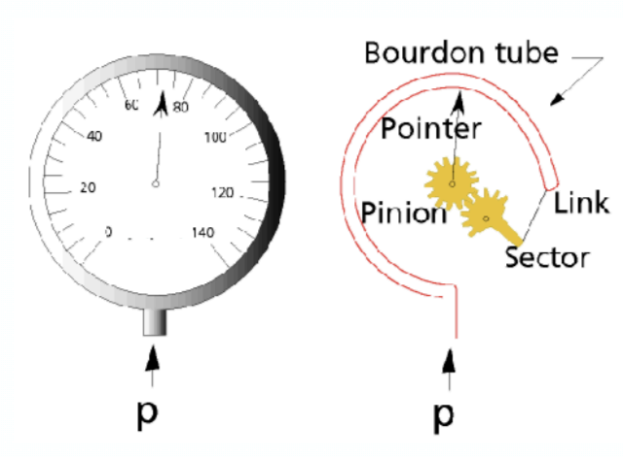
Even though manometers work with both gas and liquids, manometers are not well suited for high pressures or pressures that change rapidly over time.

Measuring pressure is also a bit more tedious as two or more liquid columns have to measured.

Bourdon Gauge

The Bourdon gauge is a pressure gauge that measures pressure due to the deformation of an elastic tube, called a bourdon tube.

When pressure in the bourdon tube increases, the tube will want to straighten out, pulling on a gear that can be translated into the rotation of a dial.

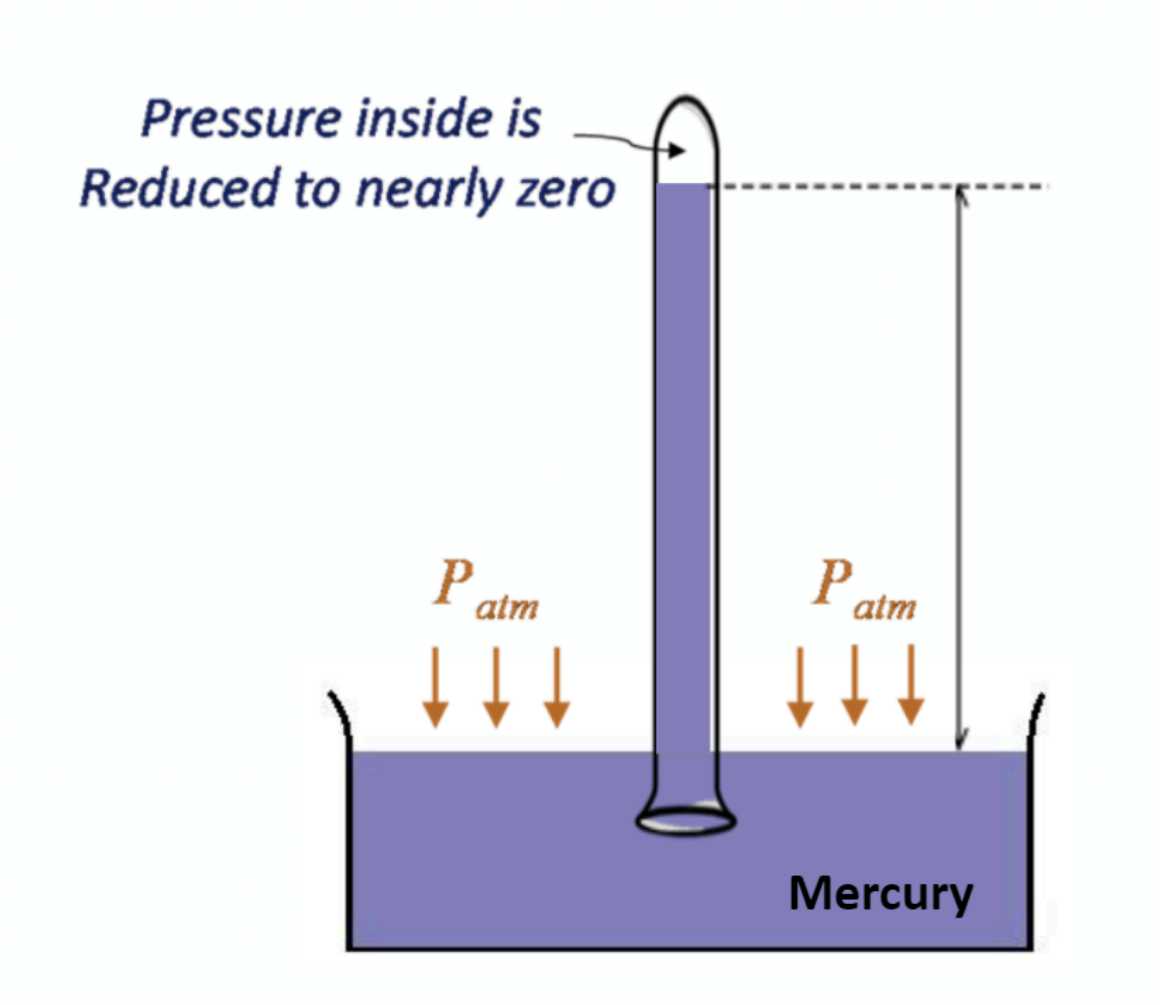


Depending on how it is calibrated, the bourdon gauge can measure absolute, gauge or vacuum pressure.

You can tell if the gauge is calibrated to measure relative to atmospheric pressure by seeing if there are negative numbers. Then, when the dial is at 0, the absolute pressure is atmospheric.

Barometers

Barometers measure the atmospheric pressure, not the fluid pressure.



barometers have the following setup; a long capped of tube with liquid immersed in a vessel of the same liquid.

The space at the end of the tube is essentially a vacuum, thus atmospheric pressure can be measured as the hydrostatic pressure due to the liquid column.